

Sound, Space and Interaction Final Assignment Report

Nervous Choir

The Creation Process

Concept

The main concept “*Nervous Choir*” tries to convey to participants is one of harmony and what happens when that harmony is broken. This was attempted by having multiple speakers interacting and singing with each other in peace when no external input is received. Consequently, this meant that the work needed to stand on its own feet if it was simply gazed at, rather than interacted with.

Once a participant does decide to interact, they will inevitably disturb the attained harmony in some way, be it distorting and interrupting the speakers’ singing, or more severely, “killing” a speaker by cutting its wiring. Depending on participant, this would either discourage them from interacting further in hopes of restoring harmony or encourage them to find other ways of messing with the speakers.

Development

Development was quite difficult, as we realized that our initial goals were a bit too ambitious. This led to us having to reevaluate what it was exactly that needed to be included in the final work and what we should cut out due to technical and time constraints. For example, our first vision of “*Nervous Choir*” included six speakers, rather than four. There were also numerous occasions when we would have differing opinions on how certain parts of the work should be realised/highlighted. While we always were able to find common ground and work together, this did take up time that was sorely needed for the realisation of the project.

Realisation

When it came to actually bringing our ideas to life, we had a mix of successes and challenges. To start with the former, communication between all the speakers was achieved nicely with any combination of speakers being supported during the so-called “searching phase”. Furthermore, although implementation of this feature was extremely challenging, we were able to have scared speakers speed up to go out of sync with the rest, before syncing up once again after a certain amount of time.

The difficulties we faced, however, far outnumbered our successes. The ultrasonic distance sensors, for example, were not working properly in our first setup due to interference between the different sensors. This meant that we had to change the

placement of the speakers from being at the corners of the room to the centre of the room with the sensors facing outwards.

Who did what

During the early stages of the project, Pure Data coding was primarily done by Payman while Mats was focusing on the Arduino hardware and code. However, we soon started to both work on Pure Data at the same time with Payman handling the communication between speakers, their vocabulary and singing, while Mats was in charge of how the speakers would react to outside input. Afterwards, we both spent time wiring everything together and tweaking the Arduino code. During a week when Payman was unable to work much on the project, Mats testing the final work mostly on his own.

The Sonic Language

We took a more literal approach to developing our own sonic language by assigning meanings to the certain sounds that the speakers made in order to make it feel as if they were communicating with a vocabulary unique to them. Inspiration was taken from how [JBL speakers sound](#) when connecting the smart devices over Bluetooth.

All the speakers also have a specific waveform assigned to them to give them a distinct “voice”. These were simple sine, triangle, sawtooth, and square waveforms, as to make the speakers sound as robotic and unnatural as possible. Initially, we planned on having all the speakers sound like a specific instrument (e.g., speaker 1 would only play guitar sounds while speaker 2 would sound like a piano). However, because we did not want the speakers to sound too “human”, we ended up going for the more computer-like waveforms.

The Interaction

Participants were able to interact with the speakers in three main ways. The first, and most prevalent manner of interaction, was facilitated by the four ultrasonic distance sensors. Should a participant come too close to any speaker, its voice would sound more distorted and slowly go out of sync with the other speakers. If this “fear state” would go on for too long, the speaker would get frustrated and yell out to the participant in beeping noises to show its frustration.

The second and third manner of interaction are closely linked to each other. Next to the speakers, a potentiometer was placed which participants could use to rate their satisfaction level regarding the speakers’ performance. Should this level get too low, a pair of scissors would be revealed which could be used to cut an antenna placed on the speakers head. If this antenna were to be cut, the speaker would “die” and emit a death screech, never producing any sound again. The other speakers would, in turn, react to this and start panicking, made audible by rapid, high-pitched beeping noises. After a certain amount of time, the remaining speakers would come together once more and

sing with the “dead” speaker remaining quiet. In this way, the speakers interacted with the participants through their audible reactions to the participants’ actions, perhaps influencing them by making them feel bad about what had occurred.

The Use of Space

It was very important to us to allow participants the option to simply observe the speakers singing in harmony without killing or scaring them. For that reason, a relatively large space was needed to make sure that participants who do not want to disturb the speakers do not accidentally do so. Furthermore, the large space meant that we were able to use the ultrasonic distance sensors to measure the distance between the speakers and the participants more gradually.

Reflection

While we are proud of the end result, many desired functions had to be cut, and the entire work had to be scaled down in order to make sure we did not miss the deadline. We severely underestimated the amount of time and effort it would take to complete the work, and therefore had long, stressful days working on it. Should we ever make another interactive sound installation in the future, it would be wise to focus on more simplistic ways to convey certain emotions and concepts, as that can be just as captivating an experience. Because the complexity of “*Nervous Choir*” made the work hard to understand for participants, we should focus on more simplistic ways to convey certain emotions and concepts to audiences.

Important

In order to run the code properly, the “arduinotest” and “Test1” patches must be opened simultaneously.